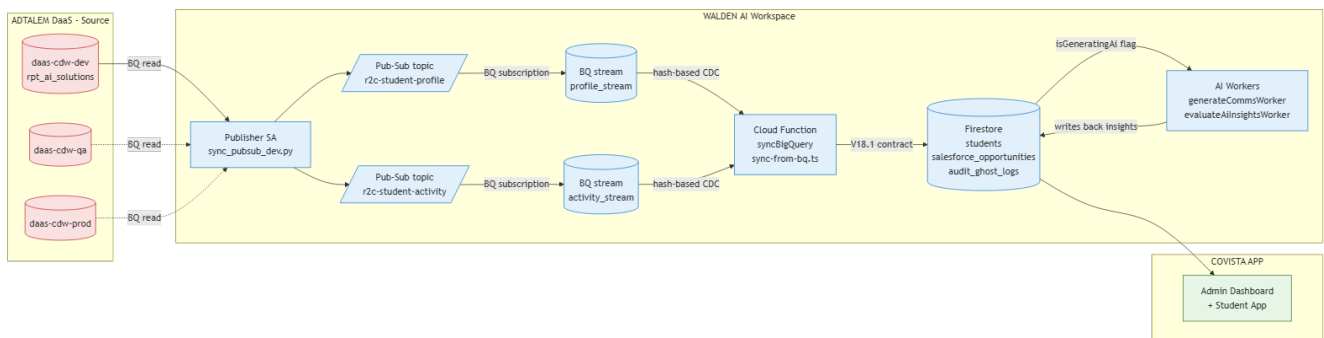


Covista R2C Pipeline — End-to-End Architecture (May 11) — Expanded

CDW → Pub/Sub → BigQuery → Firestore → Covista App, plus AI workers.

This is the long-form companion to `pipeline_architecture_may11.md`. Each section expands one hop with mechanism, schema, owner, current status, and the open gaps we need to close before prod.

1. Diagram



2. Hop-by-hop walkthrough

Hop 1 — CDW → Pub/Sub (publisher SA)

- **Source:** `daas-cdw-{dev, qa, prod}` BigQuery datasets, table family `rpt_ai_solutions.t_wldn_r2c_student_profile` and `rpt_ai_solutions.t_wldn_r2c_student_activity` (AdtaleM-owned project).
- **Mechanism:** scheduled Python publisher (`python-agent/pubsub/sync_pubsub_dev.py`) that does a bounded BigQuery read, JSON-encodes each row, and publishes to the matching Walden Pub/Sub topic.
- **Cadence:** `dev/qa` = on-demand for now (manual `python ... publish --limit N`); `prod` target = every 15 min via Cloud Scheduler.
- **Volumes verified 5/11 from Cloud Shell as `d51029691-c@mail.waldenu.edu`:**
 - `daas-cdw-dev` → 5,583 rows
 - `daas-cdw-qa` → 2 rows
 - `daas-cdw-prod` → 5,582 rows
- **Topics (Walden project `dev-wu-agenticaai-app-proj`):**
 - `r2c-student-profile`
 - `r2c-student-activity`
- **Owner:** Nagendra (publisher script, schema, contract alignment).

Hop 2 — Pub/Sub → BigQuery (native subscription)

- **Mechanism:** native **BigQuery subscription** on each topic — no code, managed by Pub/Sub itself.
- **Sink tables (Walden `covista_demo` dataset):**

 - `profile_stream` ← `r2c-student-profile`
 - `activity_stream` ← `r2c-student-activity`

- **Schema:** raw payload + `publish_time`, `subscription_name`, `message_id`, `attributes` (Pub/Sub default BQ schema).
- **Why this hop exists:** lets us replay, audit, and do CDC against the previous landed row instead of hitting CDW every time.
- **Owner:** Nagendra (terraform / IaC for the subscription).

Hop 3 — BigQuery → Firestore (**syncBigQuery** Cloud Function)

- **Code:** `functions/src/sync-from-bq.ts`.
 - `syncBigQuery()` — core sync, currently bounded to **3 pilot IDs** (A00302996, A00437050, A00409782) plus 22 Reserved/future-start students so we can validate without flooding Firestore.
 - `syncBigQueryNative` — `onCall` HTTPS entry point used by the admin dashboard "Force Sync" button.
 - `onBqSyncTrigger` — `onDocumentWritten` on `system_config/bq_sync_trigger` so an admin can flip a flag and re-run sync without redeploying.
- **Project / dataset / collection:**
 - Project: `dev-wu-agenticaai-app-proj`
 - Dataset: `covista_demo`
 - Firestore collections written:
 - `students`
 - `salesforce_opportunities`
 - `audit_ghost_logs` (rows that fail contract validation)
- **CDC mechanism:** hash of contract-relevant fields is stored per doc; only changed rows go through to Firestore. Ghost rows (no matching opportunity) are written to `audit_ghost_logs` for AM review.
- **Contract:** V18.1 (`Datacontract/covista_data_contract_v17_9.md` is the legacy file; V18.1 lives in `context/Masterlivingdocs/`).
- **Owner:** Nagendra.

Hop 4 — Firestore → AI workers

- **Code:** `functions/src/index.ts`.
 - `generateCommsWorker` — `onMessagePublished` on `generate-comms` topic. Drafts outreach copy per student using the V18.1 contract fields.
 - `evaluateAiInsightsWorker` — `onMessagePublished` on `evaluate-ai-insights` topic. Scores risk, computes next-best-action, writes results back to the same students doc under `aiInsights.*`.
- **Trigger pattern:** the sync function sets `isGeneratingAi = true` on the student doc

and publishes a Pub/Sub message; the worker picks it up, runs the model, writes results, and clears the flag.

- **Models:** Vertex Gemini (text) + a custom evaluator prompt — owned by Jaishir.
- **Owner:** Jaishir (AI generation), Nagendra (Pub/Sub plumbing + worker scaffolding).

3. Four subscription/event hops — quick table

#	Hop	Mechanism	Owner
1	CDW → Pub/Sub	Publisher SA + scheduled BQ read in <code>python-agent/pubsub/sync_pubsub_dev.py</code>	Nagendra
2	Pub/Sub → BQ	Native BigQuery subscription (no code)	Nagendra
3	BQ → Firestore	Cloud Function <code>syncBigQuery()</code> in <code>functions/src/sync-from-bq.ts</code> ; trigger via <code>onBqSyncTrigger</code> (Firestore doc write) or scheduled cron	Nagendra
4	Firestore → AI	<code>onMessagePublished</code> workers (<code>generateCommsWorker</code> , <code>evaluateAiInsightsWorker</code>) in <code>functions/src/index.ts</code> ; AI results written back to same docs	Jaishir (AI gen) + Nagendra (PubSub workers)

4. Identity / IAM summary

Today (5/11 actual state)

Identity	CDW read (Adtalem)	Pub/Sub publish (Walden)	gcloud OAuth local
<code>d51029691-c@mail.waldenu.edu</code>	✓ dev/qa/prod	✗ (no <code>pubsub.topics.list</code>)	✗ Workspace blocks client 32555940559
<code>d51029691-c@adtalem.com</code>	✗ 403 in all 3 envs	✓ topics list (granted by Kewyn)	✗ Workspace blocks client 32555940559
<code>nagversion3@gmail.com</code>	✗	✗	✓ but no work access

Implication: no single user identity covers source-read + publish, AND neither corp identity can run `gcloud auth login` locally because both Workspaces block the SDK's OAuth client. We are running today via **Cloud Shell** (web SSO is allowed) using the Walden identity for CDW reads.

Target (prod) — dedicated SA

`pubsub-cdw-publisher@dev-wu-agenticaai-app-proj.iam.gserviceaccount.com` with:

- `roles/bigquery.dataViewer` on `rpt_ai_solutions` in `daas-cdw-{dev,qa,prod}` (granted by Adtalem AM — pending Kewyn's reply).
- `roles/bigquery.jobUser` on `dev-wu-agenticaai-app-proj` (publisher project).

- `roles/pubsub.publisher` on the two Walden topics.

The BQ → Firestore leg already runs as the Firebase Admin SA — no new IAM needed there.

5. Status snapshot (5/11)

- ✓ Pub/Sub topics + BQ subscriptions exist in `dev-wu-agenticaai-app-proj`.
 - ✓ CDW read access **verified** from Walden identity in Cloud Shell across all 3 envs.
 - ✓ `sync-from-bq.ts` deployed to dev, bounded to 3 pilot students + 22 Reserved.
 - ✓ V18.1 contract merged to `feature/apr27-checklist-fafsa-rules` (`ff7c963`).
 - ⌚ Publisher SA + key — waiting on Kewyn (Adtalem-side IAM grants).
 - ⌚ Cloud Scheduler trigger for hop 1 — not wired yet (manual runs only).
 - ⌚ Full-population sync (drop the 3-ID bound) — gated on prod IAM + load test of Firestore write fan-out.
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6. Known gotchas

- **gcloud OAuth blocked on both Workspaces.** Don't waste time retrying `gcloud auth login` locally — use Cloud Shell or an SA key.
 - **Pub/Sub IAM is on the Adtalem identity, not Walden.** A single user identity cannot drive the whole pipeline; the SA is mandatory, not optional.
 - **Mermaid in VS Code preview is 8.8.0** — avoid quoted subgraph titles, `direction TB` inside subgraphs, `{{...}}` hexagons, and `classDef ..., color:#000`. Use plain titles and per-node `style` lines.
 - **Ghost rows** (CDW row with no matching `salesforce_opportunities`) land in `audit_ghost_logs` instead of `students` — review weekly with AM.
 - **`isGeneratingAi` flag must be cleared by the worker on every exit path** (success, model error, validation error). Stuck flags = dashboard shows a permanent spinner.
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7. References

- Source: `python-agent/pubsub/sync_pubsub_dev.py`
- Sync function: `functions/src/sync-from-bq.ts`
- AI workers: `functions/src/index.ts`
- Contract: `context/Masterlivingdocs/covista_data_contract_v17_9_Latest.md` (and V18.1 successor)
- Identity matrix: `/memories/repo/covista-gcp-identities.md`
- EOD log: `/memories/session/may11-pubsub-status.md`